REINFORCEMENT LEARNING

Lecture 5 –Q-Learning

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Lecture credit: Matt Gormley

1. Recap Value Function

2. Recap Q-Function

3. Deep Q-Function

1. Recap Value Function

From previous slide (Lecture 4)

Definitions: Value function and Q-value function

Following a policy produces sample trajectories (or paths) $s_0, a_0, r_0, s_1, a_1, r_1, \ldots$

How good is a state?

The **value function** at state s, is the expected cumulative reward from following the policy from state s:

$$V^\pi(s) = \mathbb{E}\left[\sum_{t\geq 0} \gamma^t r_t | s_0 = s, \pi\right]$$

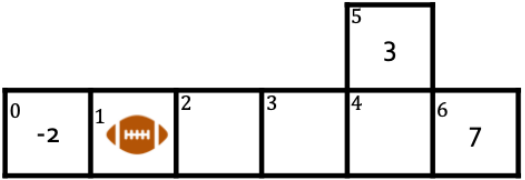
How good is a state-action pair?

The **Q-value function** at state s and action a, is the expected cumulative reward from taking action a in state s and then following the policy:

$$Q^\pi(s,a) = \mathbb{E}\left[\sum_{t\geq 0} \gamma^t r_t | s_0 = s, a_0 = a, \pi\right]$$

Value Function Example

Deterministic environment:
An action deterministically determines next state

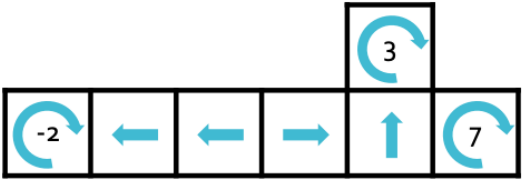


$$R(s,a) = \begin{cases} -2 \text{ if entering state 0 (safety)} \\ 3 \text{ if entering state 5 (field goal)} \\ 7 \text{ if entering state 6 (touch down)} \\ 0 \text{ otherwise} \end{cases}$$

$\gamma = 0.9$

Value Function Example

Deterministic environment:
An action deterministically determines next state



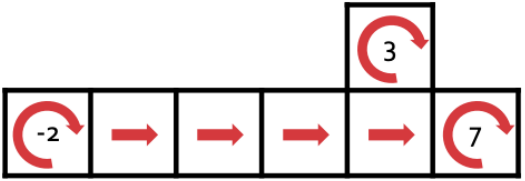
$$R(s,a) = \begin{cases} -2 \text{ if entering state 0 (safety)} \\ 3 \text{ if entering state 5 (Mield goal)} \\ 7 \text{ if entering state 6 (touch down)} \\ 0 \text{ otherwise} \end{cases}$$

$\gamma = 0.9$

				0	
0	-2	-1.8	2.7	3	0

Value Function Example

Deterministic environment:
An action deterministically determines next state



$$R(s,a) = \begin{cases} -2 \text{ if entering state 0 (safety)} \\ 3 \text{ if entering state 5 (Mield goal)} \\ 7 \text{ if entering state 6 (touch down)} \\ 0 \text{ otherwise} \end{cases}$$

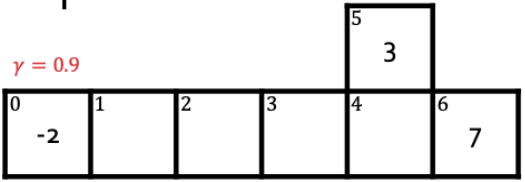
$\gamma = 0.9$

				0	
0	5.103	5.67	6.3	7	0

Value Function Example

Stochastic environment: An action stochastically determines next state

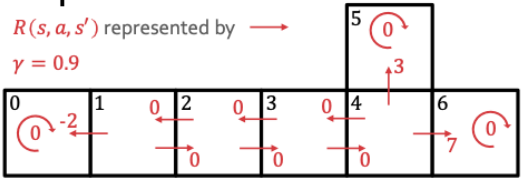
$\gamma = 0.9$



$$R(s,a,s') = \begin{cases} -2 \text{ if entering state 0 (safety)} \\ 3 \text{ if entering state 5 (Mield goal)} \\ 7 \text{ if entering state 6 (touch down)} \\ 0 \text{ otherwise} \end{cases}$$

Value Function Example

Stochastic environment: An action stochastically determines next state

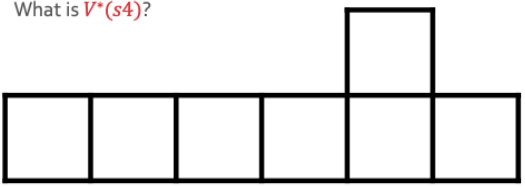


Suppose

- $p(s6 \mid s4,\ a) = 0.5$
- $p(s5 \mid s4,\ a) = 0.5$

$\gamma = 0.9$

What is $V^*(s4)$?



Bellman equation

From previous slide (Lecture 4)

The optimal Q-value function Q* is the maximum expected cumulative reward achievable from a given (state, action) pair:

$$Q^*(s,a) = \max_{\pi} \mathbb{E}\left[\sum_{t\geq 0} \gamma^t r_t | s_0 = s, a_0 = a, \pi\right]$$

Q* satisfies the following **Bellman equation**:

$$Q^*(s,a) = \mathbb{E}_{s'\sim\mathcal{E}}\left[r + \gamma \max_{a'} Q^*(s',a') | s, a\right]$$

Intuition: if the optimal state-action values for the next time-step Q*(s',a') are known, then the optimal strategy is to take the action that maximizes the expected value of $r + \gamma Q^*(s',a')$

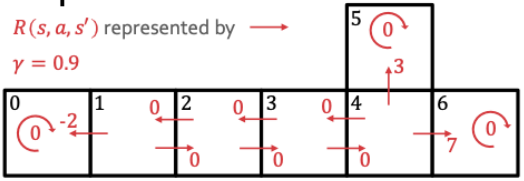
The optimal policy π* corresponds to taking the best action in any state as specified by Q*

Value Function Example

Stochastic environment: An action stochastically determines next state

$R(s, a, s')$ represented by →

$\gamma = 0.9$



Suppose

- $p(s6 \mid s4, \ a) = 0.5$
- $p(s5 \mid s4, \ a) = 0.5$

$\gamma = 0.9$

What is $V^*(s4)$? 5

V*(s)

				0	
0	3.64	4.05	4.5	5	0

2. Recap Q-Function

Optimal Q-Value

- $Q^*(s,a) = \mathbb{E}[$total discounted reward of taking action a in state s, assuming all future actions are optimal$]$

$$= R(s,a) + \gamma \sum_{s' \in \mathcal{S}} p(s' \mid s,a) V^*(s')$$

- $V^*(s') = \max_{a' \in \mathcal{A}} Q^*(s',a')$

$$Q^*(s,a) = R(s,a) + \gamma \sum_{s' \in \mathcal{S}} p(s' \mid s,a) \left[\max_{a' \in \mathcal{A}} Q^*(s',a') \right]$$

$$\pi^*(s) = \operatorname*{argmax}_{a \in \mathcal{A}} Q^*(s,a)$$

- Insight: if we know Q^*, we can compute an optimal policy π^*!

How to get optimal Q-Value?

- Algorithm 1: Online learning of Q^* (table form)
  - Inputs: discount factor γ,

    an initial state s

  - Initialize $Q(s,a) = 0 \; \forall \, s \in \mathcal{S}, a \in \mathcal{A}$
    (Q is a $|\mathcal{S}| \times |\mathcal{A}|$ table or array)
  - While TRUE, do
    - Take a random action a
    - Receive some reward $r = R(s,a)$
    - Observe the new state $s' = \delta(s,a)$
    - Update Q and s

      $$Q(s,a) \leftarrow r + \gamma \max_{a'} Q(s',a')$$

      $$s \leftarrow s'$$

Online gathering of training sample (s, a, r, s')

- Always take a random action
- Too slow convergence…

How to get optimal Q-Value?

- Algorithm 2: ϵ-greedy online learning of Q^* (table form)
  - Inputs: discount factor γ,
    an initial state s,
    greediness parameter $\epsilon \in [0, 1]$
  - Initialize $Q(s, a) = 0 \; \forall \, s \in \mathcal{S}, a \in \mathcal{A}$
    (Q is a $|\mathcal{S}| \times |\mathcal{A}|$ table or array)
  - While TRUE, do
    - With probability $1 - \epsilon$, take the greedy action $a = \underset{a' \in \mathcal{A}}{\operatorname{argmax}} \; Q(s, a')$. Otherwise (with probability ϵ), take a random action a
    - Receive reward $r = R(s, a)$
    - Observe the new state $s' = \delta(s, a)$
    - Update Q and s
      $Q(s, a) \leftarrow r + \gamma \max_{a'} Q(s', a')$
      $s \leftarrow s'$

- Sometimes, take a greedy solution (exploit what we've learned)
- Otherwise, take a random action (explore new actions)

How to get optimal Q-Value?

- Algorithm 3: ϵ-greedy online learning of Q^* (table form)
  - Inputs: discount factor γ,
    an initial state s,
    greediness parameter $\epsilon \in [0, 1]$,
    learning rate $\alpha \in [0, 1]$ ("mistrust parameter")
  - Initialize $Q(s, a) = 0 \; \forall \, s \in \mathcal{S}, a \in \mathcal{A}$
    (Q is a $|\mathcal{S}| \times |\mathcal{A}|$ table or array)
  - While TRUE, do
    - With probability $1 - \epsilon$, take the greedy action $a = \underset{a' \in \mathcal{A}}{\operatorname{argmax}} \; Q(s, a')$. Otherwise (with probability ϵ), take a random action a
    - Receive reward $r = R(s, a)$
    - Observe the new state $s' \sim p(S' \mid s, a)$
    - Update Q and s

$$Q(s,a) \leftarrow \underbrace{(1-\alpha)Q(s,a)}_{\text{Current value}} + \underbrace{\alpha\left(r + \gamma \max_{a'} Q(s',a')\right)}_{\text{Update w/ deterministic transitions}}$$

$$s \leftarrow s'$$

- Introduce learning rate
- Soft Update for Stability: Prevents the Q-table from oscillating by absorbing new experiences incrementally rather than overwriting existing knowledge.
- Averaging Stochastic Outcomes: Naturally converges to the expected value $\mathbb{E}$ over time by smoothing out random environmental noise without explicit probability calculations.

How to get optimal Q-Value?

- Algorithm 3: ϵ-greedy online learning of Q^* (table form)
  - Inputs: discount factor γ,
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    - Receive reward $r = R(s, a)$
    - Observe the new state $s' \sim p(S' \mid s, a)$
    - Update Q and s

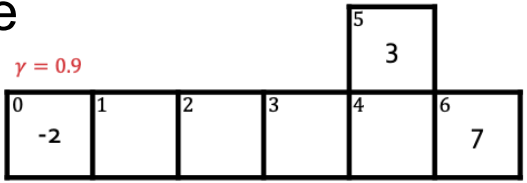
$$Q(s,a) \leftarrow \underbrace{Q(s,a)}_{\text{Current value}} + \alpha \overbrace{\Big(\underbrace{r + \gamma \max_{a'} Q(s',a')}_{\text{Temporal difference target}} - Q(s,a)\Big)}^{\text{Temporal difference}}$$

$$s \leftarrow s'$$

- Introduce learning rate
- Soft Update for Stability: Prevents the Q-table from oscillating by absorbing new experiences incrementally rather than overwriting existing knowledge.
- Averaging Stochastic Outcomes: Naturally converges to the expected value $\mathbb{E}$ over time by smoothing out random environmental noise without explicit probability calculations.

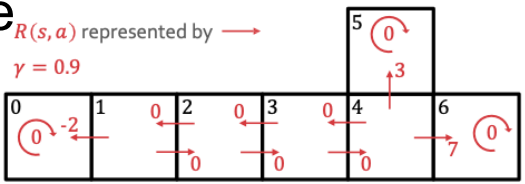
Q-Value Example

$\gamma = 0.9$



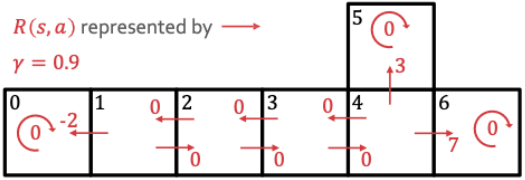
$$R(s,a) = \begin{cases} -2 \text{ if entering state 0 (safety)} \\ 3 \text{ if entering state 5 (Mield goal)} \\ 7 \text{ if entering state 6 (touch down)} \\ 0 \text{ otherwise} \end{cases}$$

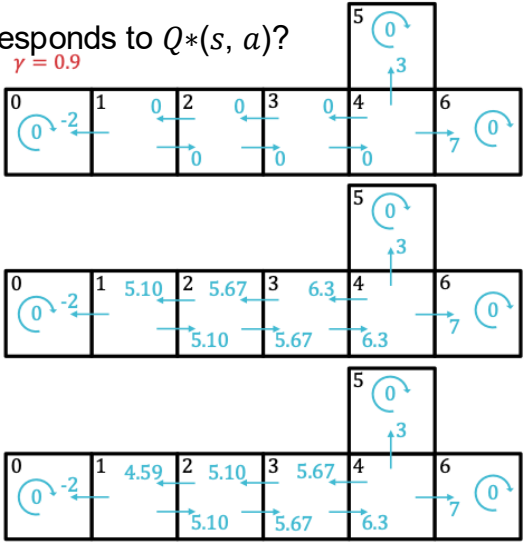
Q-Value Example



Q-Value Example

Which set of blue arrows (roughly) corresponds to $Q*(s, a)$?





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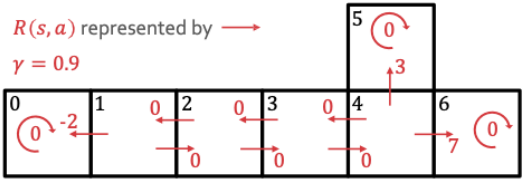
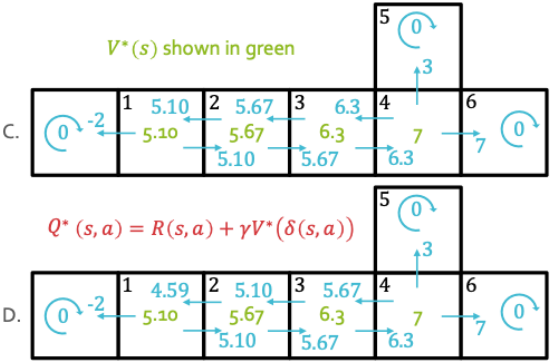
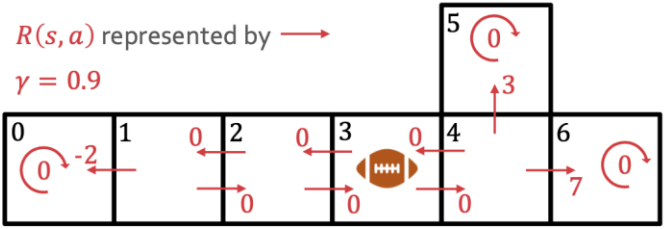




Q-Value Example

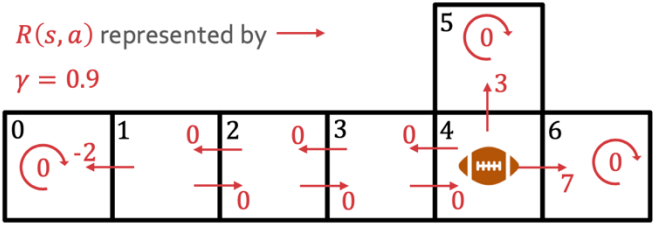


$Q(s,a)$	→	←	↑	↻
0	0	0	0	0
1	0	0	0	0
2	0	0	0	0
3	0	0	0	0
4	0	0	0	0
5	0	0	0	0
6	0	0	0	0

Q-Value Example

$R(s,a)$ represented by $\longrightarrow$

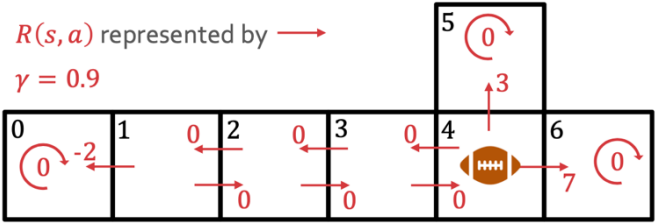
$\gamma = 0.9$



$$Q(3,\rightarrow) \leftarrow 0 + (0.9) \max_{a' \in \{\rightarrow,\leftarrow,\uparrow,\circlearrowright\}} Q(4,a') = 0$$

$Q(s,a)$	$\rightarrow$	$\leftarrow$	$\uparrow$	$\circlearrowright$
0	0	0	0	0
1	0	0	0	0
2	0	0	0	0
3	0	0	0	0
4	0	0	0	0
5	0	0	0	0
6	0	0	0	0

Q-Value Example

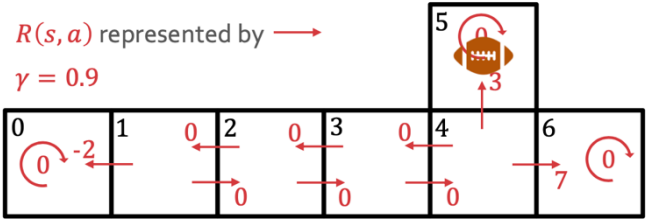


$Q(s,a)$	→	←	↑	↻
0	0	0	0	0
1	0	0	0	0
2	0	0	0	0
3	0	0	0	0
4	0	0	0	0
5	0	0	0	0
6	0	0	0	0

Q-Value Example

$R(s,a)$ represented by $\longrightarrow$

$\gamma = 0.9$



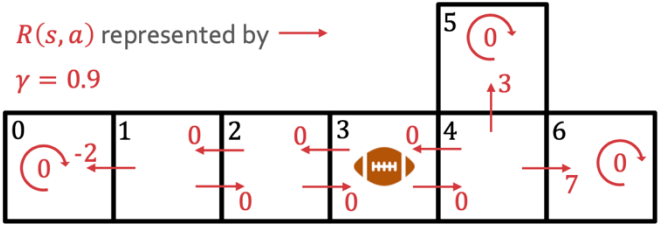
$$Q(4,\uparrow) \leftarrow 3 + (0.9) \max_{a' \in \{\rightarrow,\leftarrow,\uparrow,\circlearrowright\}} Q(5,a') = 3$$

$Q(s,a)$	$\rightarrow$	$\leftarrow$	$\uparrow$	$\circlearrowright$
0	0	0	0	0
1	0	0	0	0
2	0	0	0	0
3	0	0	0	0
4	0	0	0	0
5	0	0	0	0
6	0	0	0	0

Q-Value Example

$R(s,a)$ represented by $\longrightarrow$

$\gamma = 0.9$



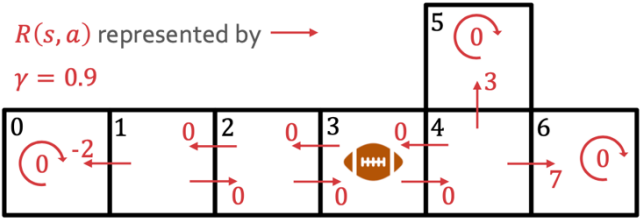
$$Q(3,\rightarrow) \leftarrow 0 + (0.9) \max_{a' \in \{\rightarrow,\leftarrow,\uparrow,\circlearrowright\}} Q(4,a') = 2.7$$

$Q(s,a)$	$\rightarrow$	$\leftarrow$	$\uparrow$	$\circlearrowright$
0	0	0	0	0
1	0	0	0	0
2	0	0	0	0
3	0	0	0	0
4	0	0	3	0
5	0	0	0	0
6	0	0	0	0

Q-Value Example

$R(s,a)$ represented by $\longrightarrow$

$\gamma = 0.9$



$$Q(3,\rightarrow) \leftarrow 0 + (0.9) \max_{a' \in \{\rightarrow,\leftarrow,\uparrow,\circlearrowright\}} Q(4,a') = 2.7$$

$Q(s,a)$	$\rightarrow$	$\leftarrow$	$\uparrow$	$\circlearrowright$
0	0	0	0	0
1	0	0	0	0
2	0	0	0	0
3	2.7	0	0	0
4	0	0	3	0
5	0	0	0	0
6	0	0	0	0

3. Deep Q-Function

Solving for the optimal policy: Q-learning

Q-learning: Use a function approximator to estimate the action-value function

$$Q(s, a; \theta) \approx Q^*(s, a)$$

function parameters (weights)

If the function approximator is a deep neural network => **deep q-learning**!

Solving for the optimal policy: Q-learning

Remember: want to find a Q-function that satisfies the Bellman Equation:

$$Q^*(s,a) = \mathbb{E}_{s' \sim \mathcal{E}}\left[r + \gamma \max_{a'} Q^*(s',a') | s,a\right]$$

Forward Pass

Loss function: $L_i(\theta_i) = \mathbb{E}_{s,a\sim\rho(\cdot)}\left[(y_i - Q(s,a;\theta_i))^2\right]$

where $y_i = \mathbb{E}_{s'\sim\mathcal{E}}\left[r + \gamma \max_{a'} Q(s',a';\theta_{i-1}) | s,a\right]$

Target is also from network's output! (target is detached and use weights of previous iterations to 'fix' the target)

Backward Pass

Gradient update (with respect to Q-function parameters θ):

$$\nabla_{\theta_i} L_i(\theta_i) = \mathbb{E}_{s,a\sim\rho(\cdot);s'\sim\mathcal{E}}\left[r + \gamma \max_{a'} Q(s',a';\theta_{i-1}) - Q(s,a;\theta_i))\nabla_{\theta_i} Q(s,a;\theta_i)\right]$$

Solving for the optimal policy: Q-learning

- Using previous iteration's outputs for the target?
- How about in the early training stage with garbage values from randomly initialized Q-function network?

Solving for the optimal policy: Q-learning

- True Reward (r) as an Anchor: The actual environment reward provides a ground-truth signal, gradually correcting the inaccurate max Q' estimates.
- High Initial Exploration (eps close to 1.0): The agent mostly ignores the untrained network and takes random actions to discover true rewards.
- Replay Buffer Warm-up: Network updates are delayed until the memory collects enough diverse transitions using pure random actions.

[Mnih et al. NIPS Workshop 2013; Nature 2015]

Case Study: Playing Atari Games

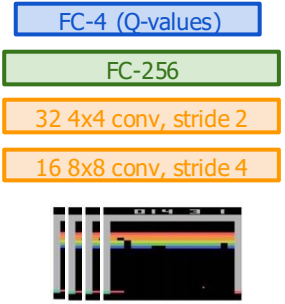
Objective: Complete the game with the highest score

State: Raw pixel inputs of the game state
Action: Game controls e.g. Left, Right, Up, Down
Reward: Score increase/decrease at each time step

[Mnih et al. NIPS Workshop 2013; Nature 2015]

Q-network Architecture

$Q(s, a; \theta)$:
neural network
with weights θ



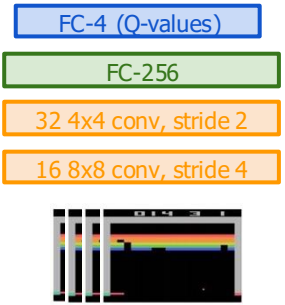
Current state s_t: 84x84x4 stack of last 4 frames
(after RGB->grayscale conversion, downsampling, and cropping)

[Mnih et al. NIPS Workshop 2013; Nature 2015]

Q-network Architecture

$Q(s, a; \theta)$:
neural network
with weights θ

A single feedforward pass to compute Q-values for all actions from the current state => efficient!



Last FC layer has 4-d output (if 4 actions), corresponding to $Q(s_t, a_1)$, $Q(s_t, a_2)$, $Q(s_t, a_3)$, $Q(s_t, a_4)$

Number of actions between 4-18 depending on Atari game

Input: state s_t

Current state s_t: 84x84x4 stack of last 4 frames
(after RGB->grayscale conversion, downsampling, and cropping)

[Mnih et al. NIPS Workshop 2013; Nature 2015]

Training the Q-network: Experience Replay

Learning from batches of consecutive samples is problematic:

- To obtain states, the agent must play through the environment, which produces sequences of highly similar, temporally correlated states.
- Samples are correlated => inefficient learning
- Current Q-network parameters determines next training samples (e.g. if maximizing action is to move left, training samples will be dominated by samples from left-hand size) => can lead to bad feedback loops

Address these problems using **experience replay**

- Continually update a **replay memory** table of transitions (s_t, a_t, r_t, s_{t+1}) as game (experience) episodes are played
- Train Q-network on random minibatches of transitions from the replay memory, instead of consecutive samples

Each transition can also contribute to multiple weight updates => greater data efficiency

[Mnih et al. NIPS Workshop 2013; Nature 2015]

Putting it together: Deep Q-Learning with Experience Replay

Algorithm 1 Deep Q-learning with Experience Replay

Initialize replay memory $\mathcal{D}$ to capacity N
Initialize action-value function Q with random weights
for episode $= 1, M$ **do**
  Initialise sequence $s_1 = \{x_1\}$ and preprocessed sequenced $\phi_1 = \phi(s_1)$
  for $t = 1, T$ **do**
    With probability ϵ select a random action a_t
    otherwise select $a_t = \max_a Q^*(\phi(s_t), a; \theta)$
    Execute action a_t in emulator and observe reward r_t and image x_{t+1}
    Set $s_{t+1} = s_t, a_t, x_{t+1}$ and preprocess $\phi_{t+1} = \phi(s_{t+1})$
    Store transition $(\phi_t, a_t, r_t, \phi_{t+1})$ in $\mathcal{D}$
    Sample random minibatch of transitions $(\phi_j, a_j, r_j, \phi_{j+1})$ from $\mathcal{D}$
    Set $y_j = \begin{cases} r_j & \text{for terminal } \phi_{j+1} \\ r_j + \gamma \max_{a'} Q(\phi_{j+1}, a'; \theta) & \text{for non-terminal } \phi_{j+1} \end{cases}$
    Perform a gradient descent step on $(y_j - Q(\phi_j, a_j; \theta))^2$ according to equation 3
  end for
end for

[Mnih et al. NIPS Workshop 2013; Nature 2015]

Putting it together: Deep Q-Learning with Experience Replay

Algorithm 1 Deep Q-learning with Experience Replay

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for episode $= 1, M$ **do**
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  for $t = 1, T$ **do**
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    Perform a gradient descent step on $(y_j - Q(\phi_j, a_j; \theta))^2$ according to equation 3
  end for
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Initialize replay memory, Q-network

[Mnih et al. NIPS Workshop 2013; Nature 2015]

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        Perform a gradient descent step on $(y_j - Q(\phi_j, a_j; \theta))^2$ according to equation 3
    end for
end for

Play M episodes (full games)

[Mnih et al. NIPS Workshop 2013; Nature 2015]

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    Perform a gradient descent step on $(y_j - Q(\phi_j, a_j; \theta))^2$ according to equation 3
  end for
end for

Initialize state (starting game screen pixels) at the beginning of each episode

[Mnih et al. NIPS Workshop 2013; Nature 2015]

Putting it together: Deep Q-Learning with Experience Replay

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    Perform a gradient descent step on $(y_j - Q(\phi_j, a_j; \theta))^2$ according to equation 3
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end for

For each timestep t of the game

[Mnih et al. NIPS Workshop 2013; Nature 2015]

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    Perform a gradient descent step on $(y_j - Q(\phi_j, a_j; \theta))^2$ according to equation 3
  end for
end for

With small probability, select a random action (explore), otherwise select greedy action from current policy

[Mnih et al. NIPS Workshop 2013; Nature 2015]

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    Perform a gradient descent step on $(y_j - Q(\phi_j, a_j; \theta))^2$ according to equation 3
  end for
end for

Take the action (a_t), and observe the reward r_t and next state s_{t+1}

[Mnih et al. NIPS Workshop 2013; Nature 2015]

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    Perform a gradient descent step on $(y_j - Q(\phi_j, a_j; \theta))^2$ according to equation 3
  end for
end for

Store transition in replay memory

[Mnih et al. NIPS Workshop 2013; Nature 2015]

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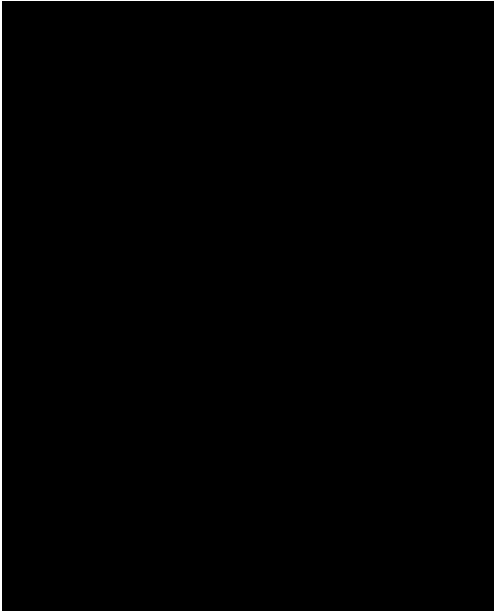
    Perform a gradient descent step on $(y_j - Q(\phi_j, a_j; \theta))^2$ according to equation 3
  end for
end for

Experience Replay: Sample a random minibatch of transitions from replay memory and perform a gradient descent step

https://www.youtube.com/watch?v=V1eYniJ0Rnk